

LLM & AI Vocabulary Sheet

Quick-revision reference for the concepts covered in Modules 1–5.

How to use: Read the term first, then the one-line explanation. Use this as a refresher before interviews, practice, or building an AI project.

1. AI / Artificial Intelligence

Software systems that perform tasks normally associated with human intelligence, such as understanding language, reasoning, and pattern recognition.

2. Generative AI

AI that generates new content such as text, code, images, audio, or video.

3. LLM / Large Language Model

An AI model trained on large amounts of text to understand and generate human-like language.

4. Prompt

The instruction or input given to an LLM to tell it what to do.

5. Prompt Engineering

Designing prompts to make LLM responses more accurate, consistent, and useful.

6. Token

A small unit of text processed by an LLM; it may be a word, part of a word, punctuation, or symbol.

7. Tokenizer

The component that converts text into tokens.

8. Token ID

The numeric identifier assigned to a token by the tokenizer.

9. Embedding

A numerical vector representation of data that captures semantic meaning.

10. Vector

An array of numbers representing information such as the meaning of text.

11. Vector Database

A database designed to store and search vectors using similarity rather than only exact keyword matching.

12. Semantic Search

Searching by meaning so differently worded but related queries can still match.

13. Similarity Search

Finding vectors that are closest to a query vector using a similarity measure.

14. RAG / Retrieval-Augmented Generation

A pattern where relevant external information is retrieved first and then provided to the LLM to ground its answer.

15. Chunk / Chunking

Breaking a large document into smaller pieces so relevant sections can be retrieved efficiently.

16. Inference

The process of using a trained model to generate an output from an input.

17. Neural Network

A machine-learning model made of connected layers that learns patterns from data.

18. Transformer

A neural-network architecture that uses attention mechanisms to process relationships between tokens efficiently.

19. Attention

A mechanism that lets a model focus on the most relevant parts of the input when processing each token.

20. Context

All information supplied to the model for the current generation, including instructions, messages, retrieved data, and tool results.

21. Context Window

The maximum amount of tokenized information a model can consider in one request.

22. System Message

Instructions that define the model's behavior, rules, role, or boundaries.

23. User Message

The user's input or request.

24. Assistant Message

A previous or current response generated by the AI.

25. Message Roles

Labels such as system, user, and assistant that describe the purpose of each conversation message.

26. LLM API

A programming interface that lets an application send requests to an LLM and receive generated results.

27. Model Configuration

Settings that control model behavior, such as model choice, temperature, output limits, and response format.

28. Temperature

A generation setting that influences randomness/variation; lower values are generally more deterministic.

29. Max Output Tokens

A limit on how many tokens the model can generate in its response.

30. Top-p

A sampling setting that limits token selection to a probability mass of likely candidates.

31. Streaming

Returning generated output incrementally instead of waiting for the complete response.

32. Structured Output

A response constrained to a machine-readable format such as JSON or a defined schema.

33. Schema

A formal definition of the expected structure, fields, types, and constraints of data.

34. Validation

Checking that an AI response has the expected structure, types, required fields, and business rules.

35. Tool Calling / Function Calling

A mechanism where the LLM requests a predefined function or external tool to retrieve data or perform an action.

36. Tool

A function, API, database operation, or service that an AI application makes available to the LLM.

37. Function Calling

The LLM's request to execute a particular function with specific arguments.

38. Agent

An AI system that can decide which tools or steps to use to accomplish a goal.

39. Chat History

Previously exchanged user and assistant messages stored for conversation continuity.

40. Conversation Management

Storing and organizing multi-turn messages so an application can maintain a useful conversation.

41. Context Management

Selecting and organizing the most relevant information to send to the model within its context window.

42. Short-Term Memory

Recent conversation context included directly in the current request.

43. Long-Term Memory

User or application information stored externally and retrieved when relevant.

44. Semantic Cache

A cache that reuses results for semantically similar requests using embeddings and vector similarity.

45. Redis Cache

A fast key-value store commonly used for exact-response caching, sessions, queues, and related workloads.

46. Rate Limit

A provider-imposed limit on requests or tokens within a time period.

47. RPM

Requests Per Minute; a rate-limit measure for API calls.

48. TPM

Tokens Per Minute; a rate-limit measure for token processing.

49. HTTP 429

The common HTTP status code meaning the application has exceeded a service's rate limit.

50. Exponential Backoff

A retry strategy that waits progressively longer after repeated failures.

51. Cost Awareness

Designing and operating AI systems with attention to token, model, embedding, database, and infrastructure costs.

52. Token Usage

The number of input and output tokens consumed by a request.

53. Latency

The time taken for a request or operation to produce a result.

54. Monitoring

Continuously tracking health, performance, usage, errors, cost, and quality.

55. Debugging

Finding and fixing problems in prompts, retrieval, tools, APIs, or application logic.

56. Observability

The ability to understand system behavior through logs, metrics, and traces.

57. Tracing

Recording the sequence of operations in an AI workflow so each step can be inspected.

58. Evaluation / Evals

Systematically measuring AI outputs against quality criteria or expected results.

59. Zero-Shot Prompting

Asking the model to perform a task without examples.

60. One-Shot Prompting

Providing one example before the actual task.

61. Few-Shot Prompting

Providing several examples so the model can infer the desired pattern.

62. Persona Prompting

Defining how the AI should communicate, such as friendly, formal, or concise.

63. Role-Based Prompting

Assigning the AI a role or expertise, such as React Architect or HR Assistant.

64. Prompt Design Pattern

A reusable prompt structure for common tasks, such as role + task + context + output format.

65. Delimiter

A marker used to clearly separate instructions from input data.

66. Prompt Template

A reusable prompt with placeholders filled dynamically by the application.

67. Prompt Debugging

Analyzing and improving a prompt when its output is incorrect, inconsistent, or low quality.

68. Prompt Evaluation

Testing a prompt systematically for correctness, relevance, consistency, format, and other criteria.

69. Hallucination

A confident-looking but incorrect, fabricated, or unsupported model response.

70. LangChain

An application framework/library that helps orchestrate LLMs, prompts, tools, retrieval, structured output, and agent workflows.

71. LangSmith

An observability and evaluation platform used to trace, debug, test, and evaluate LLM applications.

72. Promptfoo

A tool for automated prompt/model comparison and regression testing.

73. DeepEval

An evaluation framework for measuring LLM application quality with automated metrics.

74. OpenAI Evals

A framework for evaluating model and prompt behavior against benchmark or custom evaluation tasks.